Comparing Methods for Combining Bone Shape Models in the Tibiofemoral Joint with Knee Osteoarthritis

Edi Zhang[1], James R. Peters PhD[1], Mingrui Yang PhD[1], Xiaojuan Li PhD[1]

[1]Department of Biomedical Engineering, Cleveland Clinic, Cleveland, OH,

Background

- Shape models (SMs) derived from MRI segmentations of femur and tibia increasingly used to quantify 3D bone shapes in knee osteoarthritis (KOA)
- Bone shape shown to be associated with KOA onset and predictive of KOA development post-traumatic knee injury [1-2]
- Nearly all bone SMs created using individual bone surfaces or sets of surfaces arranged by anatomical location [2-3]
- Principal component analysis (PCA) widely used method to create SMs [3]
  - ➢ Conceptualized as fitting ellipsoid to data distribution
- Shouldn't assume distribution of bone shapes in KOA well represented by PCA features

Objectives

- Investigate different methods for generating combined SMs of KOA; namely combining surfaces and combining PC scores of individual SMs

Methods

Subject Sample

- 275 segmentations of tibia and femur from publicly available OAI ZIB data set [4]
  - ➢ 55 subjects from each Kellgren-Lawrence (KL) grade (0-4)

Femur and Tibia Surface Models

- 3D surface reconstructions of MRI segmentations using marching cubes algorithm [5]
- Surfaces smoothed using improved Laplacian smoothing [6]

Surface Registration

- Normalized subjects by centering and scaling each surface using centroid and centroid size respectively
- Generalized Procrustes analysis used to remove size and alignment variations across subjects [6]
- Extended eigenvector dimension to match surface point dimension using eigenvectors from the small (275 x 275) covariance matrix since less subjects than surface points [7]

Analysis Method

- 5-fold cross validation
  - ➢ Randomly split data into training and testing sets of 220 and 55 respectively, with equal subjects from each KL grade

Shape Models

- <u>PCA on Individual Surfaces</u>: Singular value decomposition (SVD) of individual surface point covariance matrix [3]
- <u>PCA on Concatenated Surfaces</u>: SVD of concatenated tibia and femur surface point covariance matrix
- <u>PCA on Concatenated PC Scores</u>: SVD of concatenated PC scores for individual tibia and femur SMs

Model Comparisons

- Comparison between SMs using three measures: compactness, generalizability [12], separability, and qualitative directions of variance
- <u>Compactness</u>: number of model components required to reach specified expressivity
- Evaluated using number of shape components required to reach 95% variability
- <u>Generalizability</u>: ability to extrapolate outside training set
  - ➢ Evaluated using root-mean-square-error (RMSE) of model reconstructions for excluded subjects in each fold
- <u>OA-scores</u>: measure of structural OA severity
  - ➢ OA-scores defined similarly to B-score used for the femur in KOA [1]
- <u>Separability</u>: ability to separate KL-grades of subjects
  - ➢ Evaluated using Fréchet Distance (FD) of adjacent KL-grade distributions of subject OA-scores [8]

Results

- Flexion-extension (Figure 1-a) and axial rotation (Figure 1-b) account for 32% of the variance in combined surface model
- Combined score model (Figure 1-c) and individual femur and tibia model (Figure 1-d) show dilation of femoral condyles, widening of trochlear groove and medial tibial plateau, and changes in tibial slope

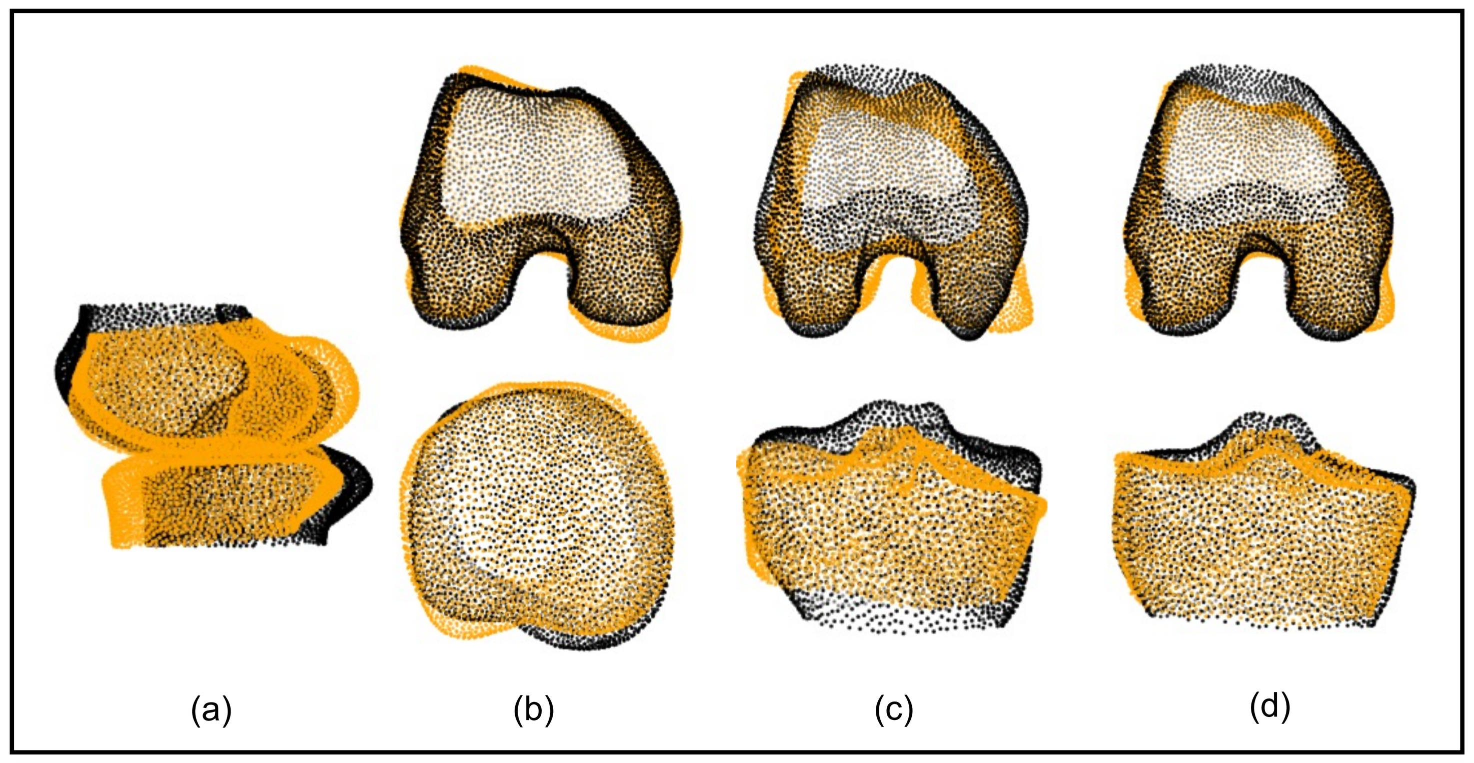

Figure 1. Black and orange figures show +3 std and -3 std from the mean along (a) PC 1 and (b) PC 2 of combined surface model, (c) PC 1 of combined score model, PC 1 of individual femur and tibia model

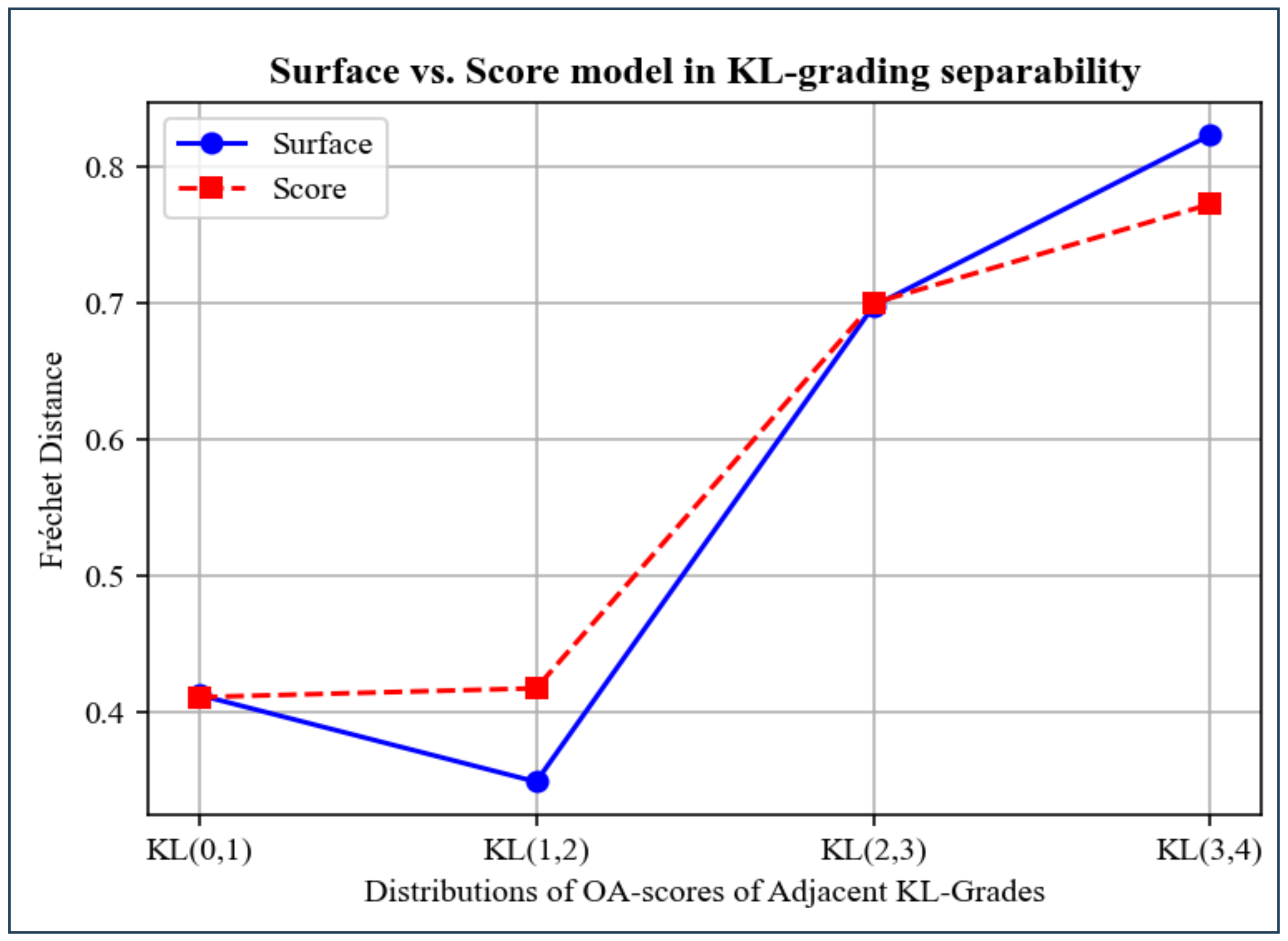

Figure 2. FD of adjacent KL-grading distributions. KL(0,1) represents the FD of the 5-fold average OA-scores of KL-0 and KL-1 (Higher FD represents better separability)

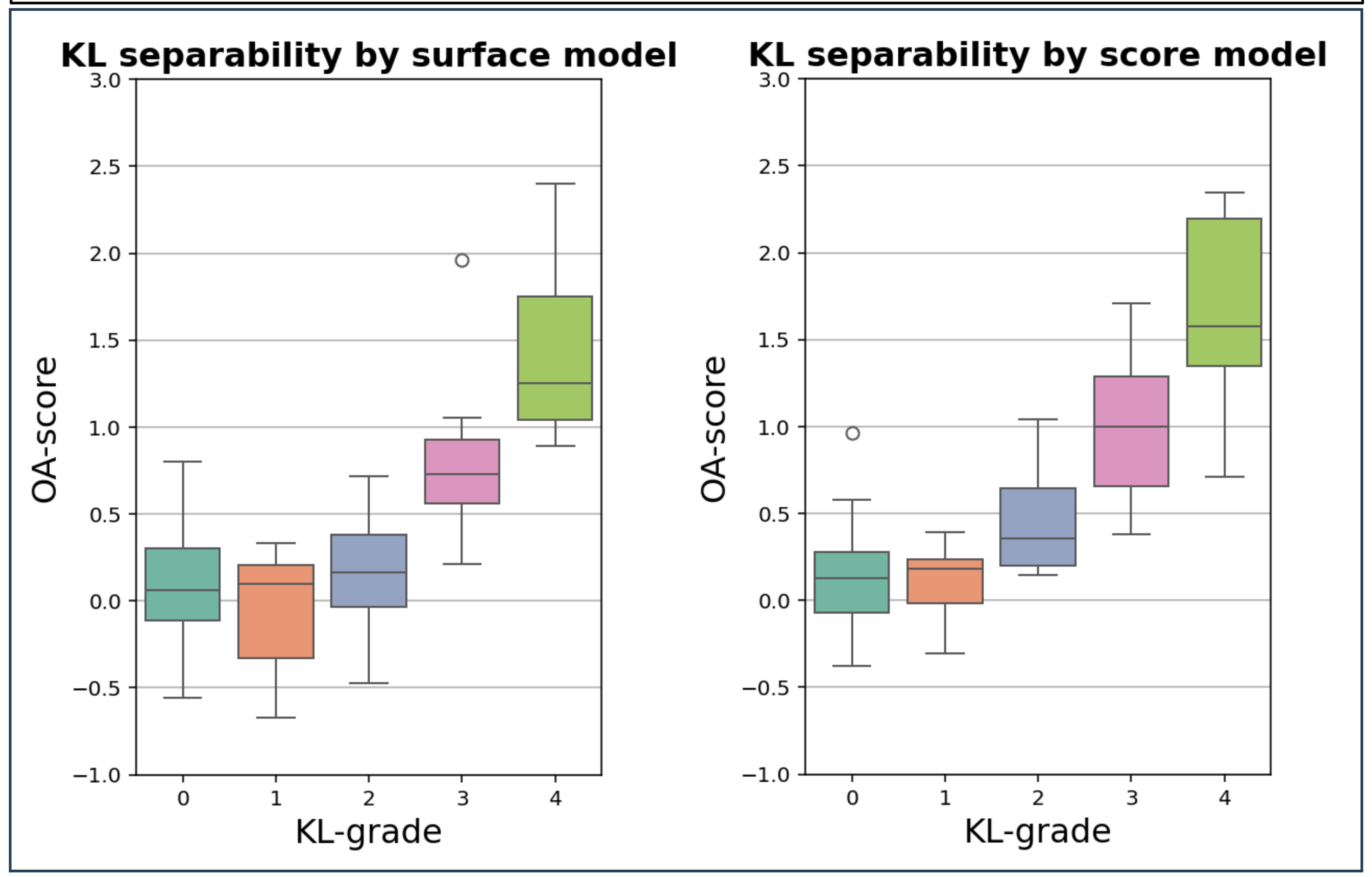

Figure 3. Subject OA-scores separated by KL grade averaged across 5 folds. KL-0 shown in turquoise, KL-1 shown in orange, KL-2 shown in blue, KL-3 shown in pink, and KL-4 in lime.

- In terms of FD, The score model separated KL-1 and KL-2 (Non-OA) and (OA) better while the surface model separated KL-3 and KL-4 (both OA) better (Figure 2)
- Score model produces higher OA-scores for KL-1 through KL-4, and surface model failed to significantly differentiate KL-0 and KL-2 (Figure 3)
- Overlapping error bars likely attributed to natural differences between subjects (Figure 3)

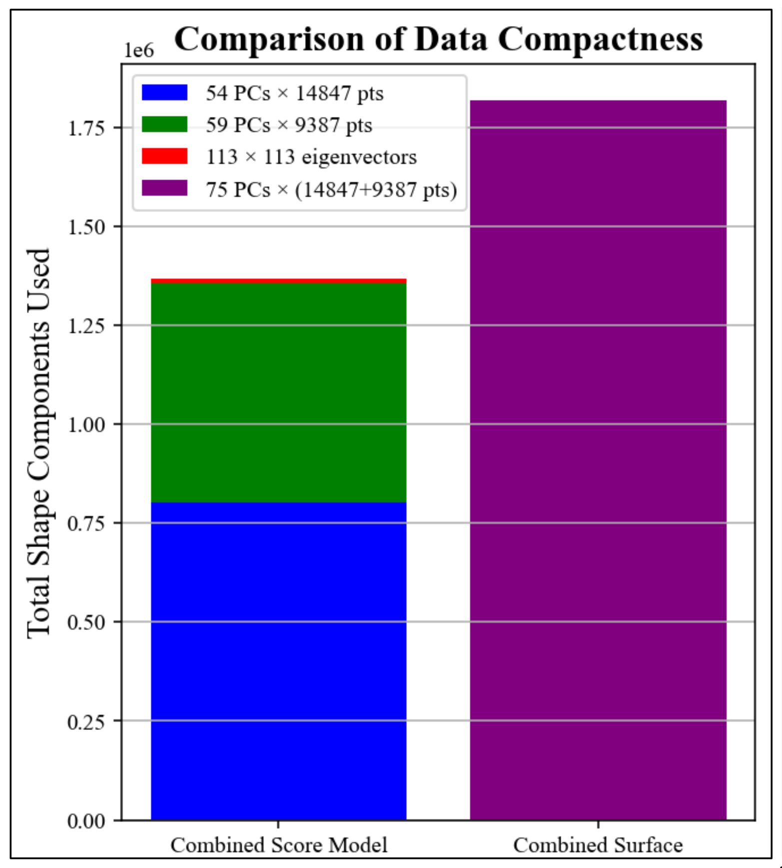

Figure 4. Number of components values used; 113x113 eigenvectors are for combined PC scores

RMSE of Surface Reconstructions

	μ (mm)
Combined Scores	0.62
Combined Surfaces	0.69
Femur	0.49
Tibia	0.43

Figure 5. Average RMSE of reconstruction error in (mm) of each model.

- Combined surface model uses about 4.2e6 more component values than combined score model (Figure 4)
- Individual femur and tibia SMs require blue and green sections respectively (Figure 4)
- Combined score model showed lower average RMSE than combined score model on 5-fold cross validation, with individual SMs having the lowest average RMSE (Figure 5)

Discussion

- Combined score model found to be more compact, better separability of KL-1 (Non-OA) and KL-2 (OA), and better reconstruction accuracy
- Separation of Non-OA grades (KL-0, KL-1) and OA (KL-2, KL-3, KL-4) is more significant in characterizing development of early-stage OA compared to separating KL-grades within OA cluster.
- About 32% of variance in combined surface model accounts for orientation differences, not useful for understanding diseases like KOA, which affects shape of bone
- RMSE difference of 0.07 mm (Figure 5) between combined score and combined surface model is not necessarily practically significant.

Conclusions

- Combined Scores model overall outperforms Combined Surfaces model for reconstruction accuracy, compactness, and separability
- Combined Scores model better characterizes shape differences as opposed to orientation differences
- Findings need to be confirmed in larger datasets

References

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